

SOC Alert Management and Incident Response Report

Submitted By:

B.PRANEETH

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INTRODUCTION:

This practical task was performed to understand the basic workflow followed in a Security Operations Center (SOC). Different activities related to alert classification, incident handling, alert triage, and investigation documentation were completed during this task. The practical work helped in understanding how SOC analysts monitor suspicious activities, classify alerts based on severity, and document incidents during investigations.

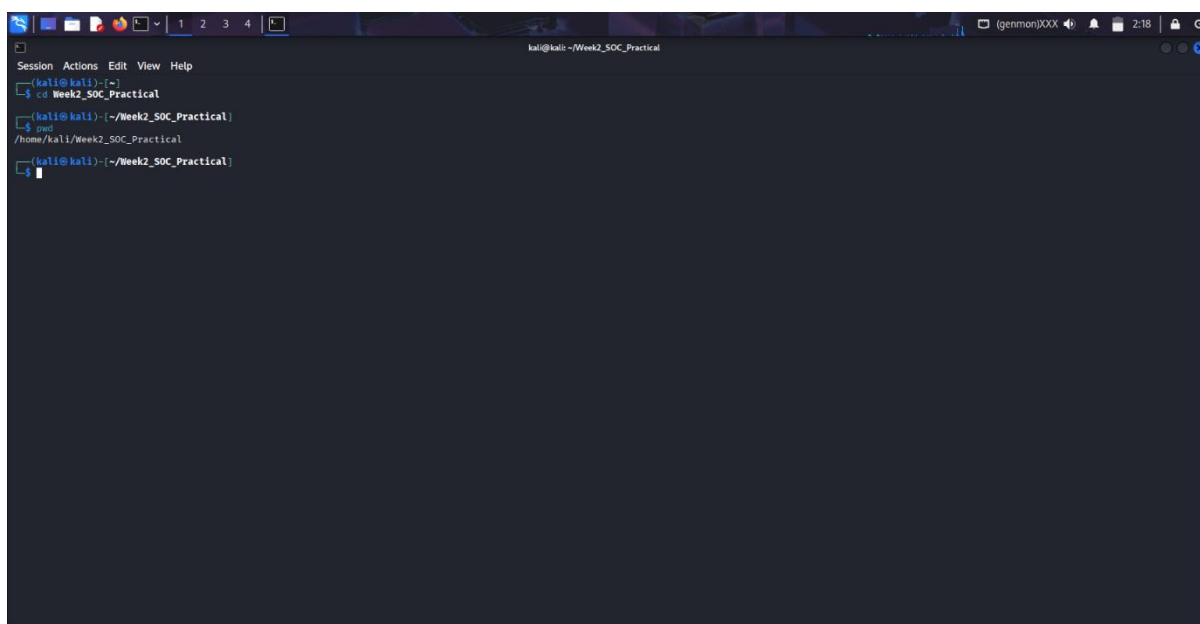
Objectives:

- To understand SOC alert classification
- To learn alert prioritization methods
- To practice incident documentation
- To understand investigation workflow
- To learn basic SOC monitoring activities

Tools Used:

- Kali Linux
- Microsoft Word
- SSH Command

- VirusTotal
- AlienVault OTX
- Wazuh
- CrowdSec
- Metasploit Framework
- GitHub



```
kali@kali: ~/Week2_SOC_Practical
Session Actions Edit View Help
kali@kali:~$ cd Week2_SOC_Practical
kali@kali:~/Week2_SOC_Practical$ pwd
/home/kali/Week2_SOC_Practical
kali@kali:~/Week2_SOC_Practical$
```

Figure 1: ENVIRONMENT SETUP

Explanation:

The environment setup process was completed in Kali Linux by creating and verifying a dedicated working directory for the SOC practical task. Terminal commands such as `cd` and `pwd` were used to confirm the correct directory location before starting the practical activities. This working environment helped organize screenshots, documentation files, and workflow-related content in a structured manner throughout the task. Maintaining a separate directory also improved file management and made the overall documentation process easier during the SOC practical implementation.

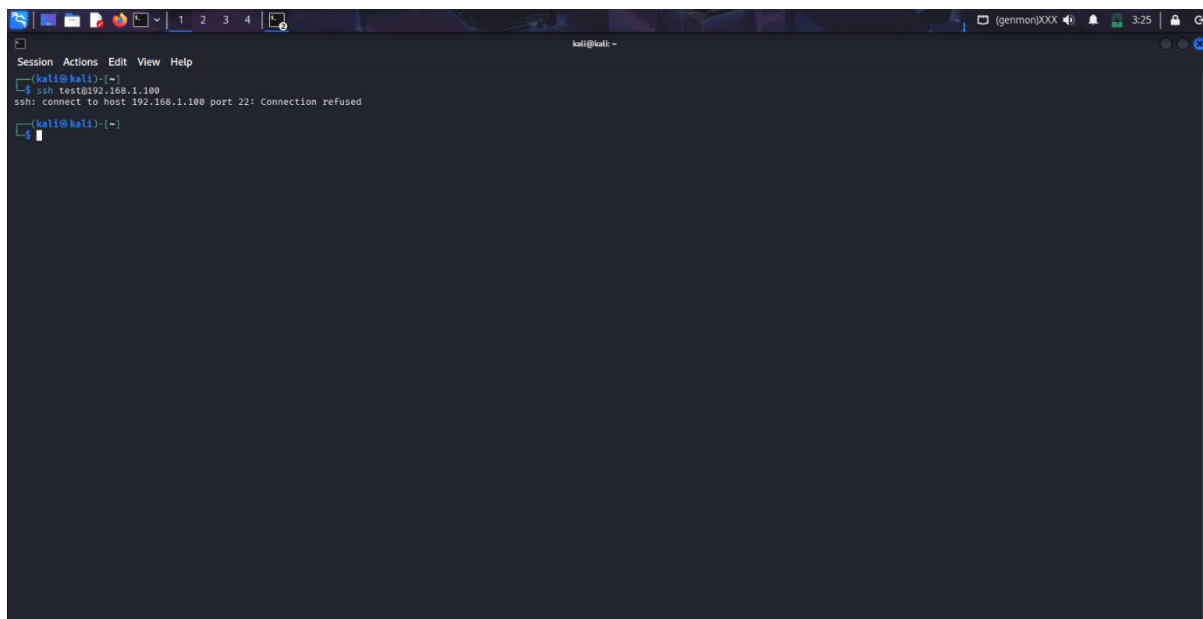


Figure 2: Failed SSH Login Attempt

Explanation:

A failed SSH login attempt was simulated using the SSH command in Kali Linux to understand how unauthorized access attempts may appear during SOC monitoring activities. During the connection process, the target system rejected the request and displayed a “Connection Refused” message. This practical activity helped in understanding how failed connection attempts can generate alerts and suspicious login events in security monitoring environments. The exercise also provided basic knowledge about SSH-related alert analysis and incident investigation procedures followed in SOC operations.

Alert Classification Table:

Alert ID	Alert Type	Priority	MITRE Technique
001	Phishing Email	High	T1566
002	Brute-force SSH	Medium	T1110
003	Port Scan	Low	T1046

Explanation:

The alert classification table was created to understand how SOC analysts categorize different security alerts based on severity and attack type. Alerts such as phishing emails, brute-force SSH attempts, and port scanning activities were assigned different priority levels according to their potential impact on the system. MITRE ATT&CK techniques were also mapped to understand attack classification methods used in SOC operations.

Incident Ticket:**Incident Title:**

Brute-force SSH Attempt

Description:

Multiple failed SSH login attempts were detected from source IP 192.168.1.100.

Priority:

Medium

Assigned To:

SOC Analyst

This incident was classified as a medium priority alert because repeated failed login attempts may indicate suspicious or unauthorized access activity.

Explanation:

An incident ticket was created to document the failed SSH login activity detected during monitoring. The ticket included details such as incident title, description, priority level, and assigned analyst information. Incident tickets help SOC teams track and manage suspicious activities in an organized manner during investigations.

Investigation Steps:

Timestamp	Action
10:00 AM	Detected failed SSH attempts
10:10 AM	Checked source IP
10:20 AM	Monitored network traffic
10:30 AM	Documented the incident

Explanation:

The investigation process was documented step-by-step to record the activities performed during alert analysis and monitoring. Different actions such as detecting failed login attempts, checking source IP addresses, monitoring traffic activity, and documenting the incident were included in the workflow.

Alert Triage Table:

Alert ID	Description	Source IP	Priority	Status
004	Brute-force SSH	192.168.1.100	Medium	Open

Explanation:

The alert triage table was created to understand how SOC analysts prioritize and manage alerts based on severity and suspicious activity. Important information such as alert description, source IP address, priority level, and alert status was documented as part of the triage process.

Threat Intelligence Validation:

Threat intelligence validation was performed to verify suspicious indicators collected during alert triage activities. The source IP address was analyzed using VirusTotal and AlienVault OTX to identify malicious reputation and suspicious activity associated with the indicator. This activity helped in understanding how SOC analysts validate Indicators of Compromise (IOCs) using external threat intelligence platforms during investigations.

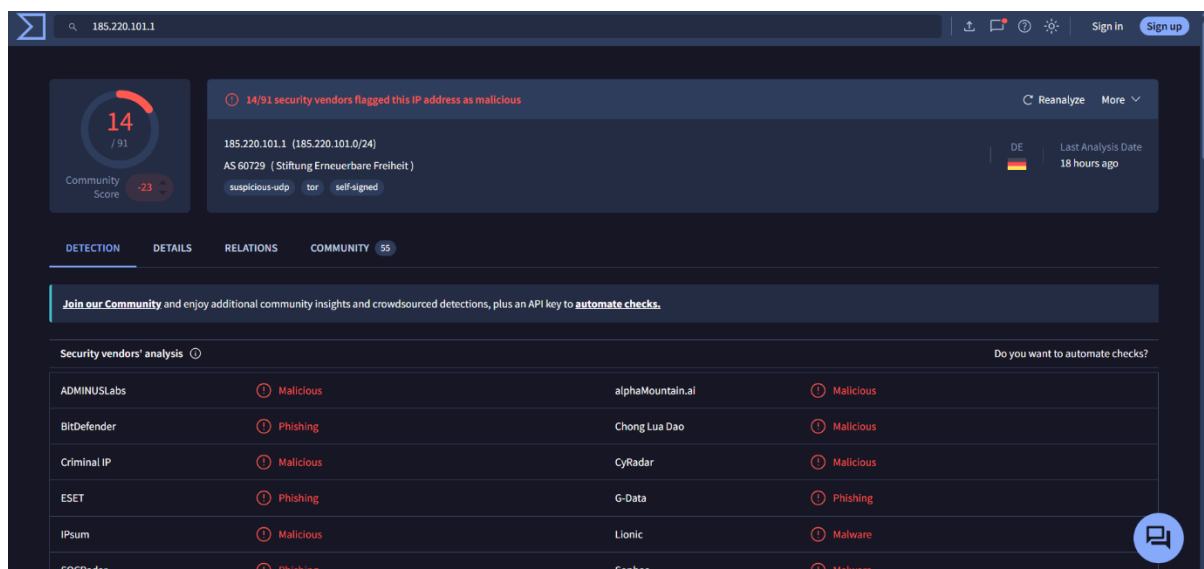


Figure 3: IOC Validation using VirusTotal

Explanation:

The suspicious IP address was validated using VirusTotal to identify malicious reputation and threat indicators associated with the alert.

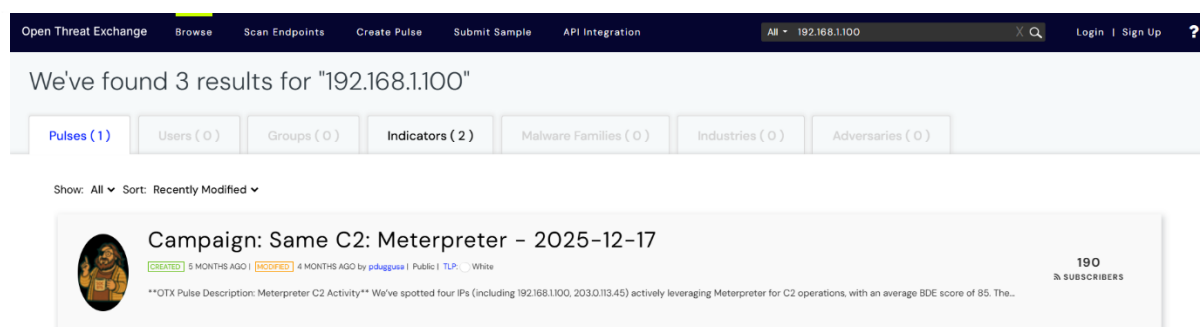


Figure 4: Threat Intelligence Analysis using AlienVault OTX

Explanation:

The alert source IP was cross-verified using AlienVault OTX to identify threat intelligence information and suspicious activity associated with the indicator. This helped in understanding how SOC analysts validate suspicious indicators during investigations.

Alert Priority Levels:

Alert	CVSS Score	Priority
Log4Shell	9.8	Critical
Phishing Email	7.5	High
Port Scan	3.0	Low

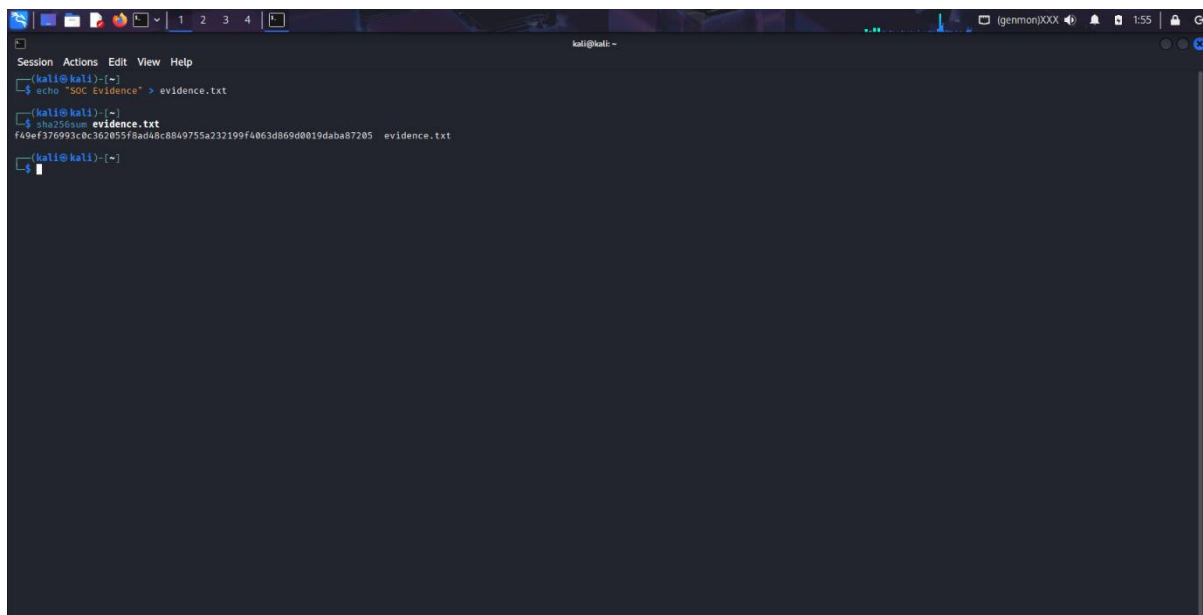
Explanation:

The alert priority level table was created to understand how different security incidents are categorized using CVSS scores and severity levels. Critical threats such as Log4Shell vulnerabilities were assigned higher priority compared to lower-risk activities like port scanning. Proper

prioritization helps SOC teams respond quickly to serious security incidents.

Evidence Preservation and Analysis:

Evidence preservation activities were performed to understand how SOC analysts maintain the integrity of collected forensic evidence during investigations. A sample evidence file was created and verified using SHA256 hashing techniques in Kali Linux. This activity helped in understanding chain-of-custody procedures and evidence integrity validation methods used during incident response operations.



```
kali@kali:~$ echo "SOC Evidence" > evidence.txt
kali@kali:~$ sha256sum evidence.txt
f49ef376993c8c362055f8ad4c8849755a232199f4063d869d0019daba87205  evidence.txt
kali@kali:~$
```

Figure 5: SHA256 Hash Verification for Evidence Integrity

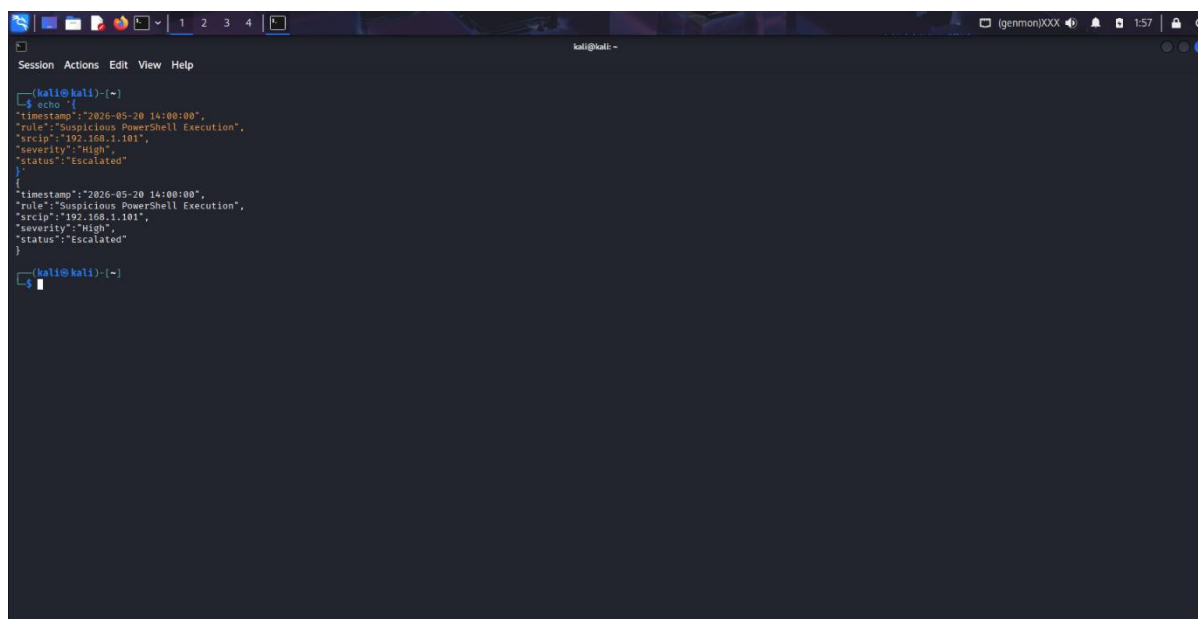
Explanation:

A SHA256 hash value was generated to verify the integrity of the collected evidence file during the investigation process.

Alert Monitoring and Detection:

A simulated Wazuh alert monitoring activity was performed to understand how SOC analysts detect suspicious activities in security monitoring environments. A mock security alert containing suspicious PowerShell execution details was generated in Kali Linux terminal output to simulate alert monitoring procedures. This activity helped in

understanding alert severity classification and escalation workflows used in SOC operations.



```

kali@kali: ~
Session Actions Edit View Help

(kali@kali)~$ echo '{
  "timestamp": "2026-05-20 14:00:00",
  "rule": "Suspicious PowerShell Execution",
  "srcip": "192.168.1.101",
  "severity": "High",
  "status": "Escalated"
}'
{
  "timestamp": "2026-05-20 14:00:00",
  "rule": "Suspicious PowerShell Execution",
  "srcip": "192.168.1.101",
  "severity": "High",
  "status": "Escalated"
}
(kali@kali)~$
  
```

Figure 6: Simulated Wazuh Security Alert Monitoring

Explanation:

A simulated security alert was generated to understand how SOC analysts monitor suspicious activities and prioritize alerts based on severity.

Response and Containment:

A simulated response and containment activity was performed to understand how suspicious IP addresses can be blocked during security incidents. A blocked IP monitoring simulation was generated to represent how CrowdSec can automatically restrict malicious traffic in SOC environments. This practical activity helped in understanding incident containment procedures followed during attack response operations.

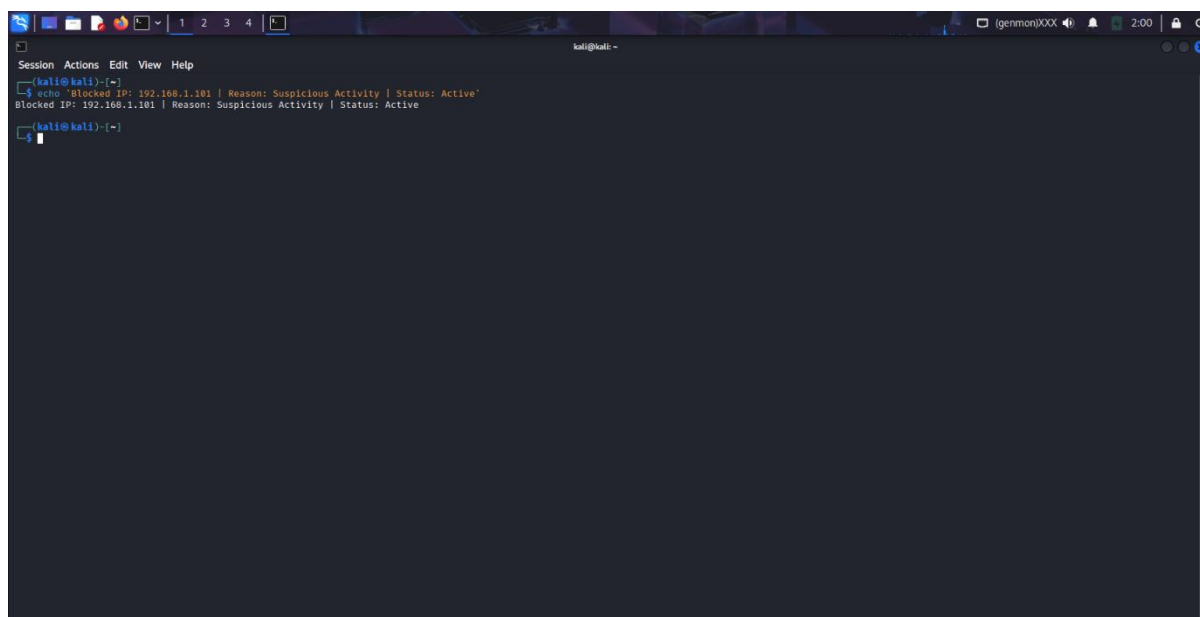


Figure 7: CrowdSec Blocked IP Monitoring

Explanation:

A simulated IP blocking activity was performed to understand how suspicious traffic can be restricted during incident response procedures.

Capstone Attack Simulation:

A basic attack simulation activity was performed using Metasploit Framework in Kali Linux to understand how SOC analysts investigate exploitation attempts in monitoring environments. The simulation demonstrated how suspicious attack behavior can generate alerts, which are later analyzed and documented during incident response procedures. This activity improved understanding of attack detection, alert triage, and SOC investigation workflows.

Figure 8: Metasploit Attack Simulation in Kali Linux

Explanation:

Metasploit Framework was used to simulate attack-related activity and understand how SOC analysts investigate suspicious exploitation attempts.

Phishing Investigation Checklist:

- ✓ Verify email headers
- ✓ Check suspicious links
- ✓ Identify affected users
- ✓ Scan attachments
- ✓ Report suspicious activity

Explanation:

The phishing investigation checklist was prepared to understand the standard verification process followed during phishing email analysis. Different validation steps such as checking sender details, suspicious

links, and email attachments were included as part of the investigation workflow.

Stakeholder Briefing:

A simulated security incident involving suspicious SSH login attempts was analyzed and documented successfully. Alert triage and incident prioritization procedures were followed during the investigation process. Threat intelligence validation and evidence verification activities were also completed to support the investigation workflow. Appropriate containment actions were simulated to reduce the impact of suspicious activity within the monitoring environment.

Observations:

- Different alert types were classified successfully.
- Failed SSH login activity was simulated and documented.
- Alert triage workflow was understood successfully.
- Incident documentation and investigation procedures were practiced.
- SOC monitoring and prioritization concepts were understood during the task.

Conclusion:

This practical task helped in understanding the workflow followed in Security Operations Centers (SOC) during alert monitoring and incident response activities. Different practical exercises related to alert classification, threat intelligence validation, evidence preservation, incident triage, and attack simulation were performed successfully. The task improved understanding of SOC investigation procedures, alert prioritization techniques, and cybersecurity documentation practices used during security monitoring operations.